

Cherenkov diffraction radiation

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This report illustrates a simple ChDR setup, the single-electron literature review, the review of finite-difference time-domain (FDTD) material interfaces, and the implications for the ChDR mini-app based on IPPL FEL. The electron remains in vacuum at a gap a from the radiator. Entering the longitudinal region beside the material is distinct from crossing into the dielectric.

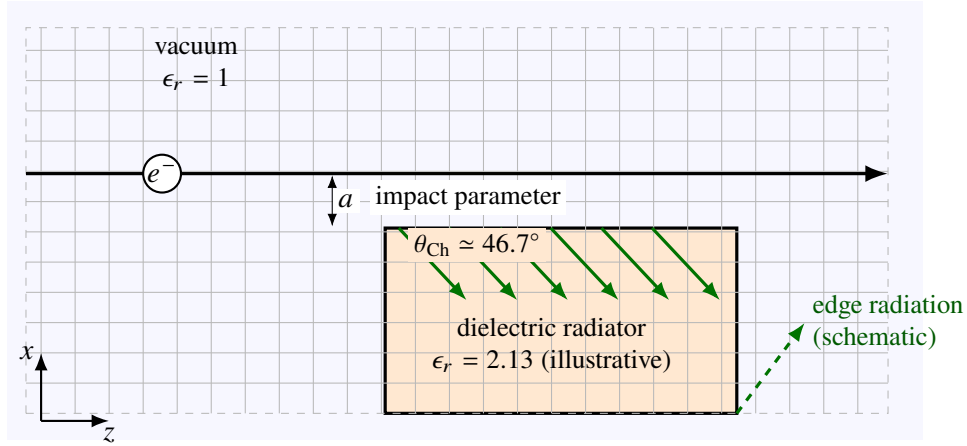


Figure 1: Schematic x - z view of the setup draft. The electron travels in vacuum along $+z$, at gap a above the finite radiator. The computational grid encompasses both materials. Solid green arrows indicate internal radiation; the dashed edge-radiation arrow is qualitative, not a prediction of the external observation angle.

1 Geometry and fixed beam parameters

Let us fix the AWA like beam parameters as follows, assuming 60 MeV means kinetic energy and 3 mm means the rms longitudinal length of a Gaussian bunch. The magnitude of the bunch charge is 5 nC. The beam travels along $+z$, with x normal to the beam-facing surface and y the other transverse coordinate. Thus $\sigma_z = 3$ mm; σ_x and σ_y denote the transverse beam sizes. The dielectric (fused silica) is initially an illustrative nonmagnetic, lossless medium with relative permittivity $\epsilon_r = 2.13$.

Quantity	Value
Electron kinetic energy K	60 MeV
Total energy per electron E_{tot}	60.511 MeV
Lorentz factor $\gamma = 1 + K/(m_e c^2)$	118.4171
Normalized speed $\beta = \sqrt{1 - \gamma^{-2}}$	0.99996434
Longitudinal rms length σ_z	3 mm
Rms duration $\sigma_t = \sigma_z/(\beta c)$	10.0073 ps
Gaussian pulse FWHM	23.6 ps
Gaussian peak current magnitude I_{peak}	199 A
Charge magnitude Q , electron number $N = Q/e$	5 nC, 3.12×10^{10}
Bunch kinetic energy NK	0.30 J
Refractive index $n = \sqrt{2.13}$	1.45945

The duration, peak current and energy follow from

$$t_{\text{FWHM}} = 2\sqrt{2 \ln 2} \sigma_t, \quad I_{\text{peak}} = \frac{Q}{\sqrt{2\pi} \sigma_t}, \quad (1.1)$$

$$E_{\text{tot}} = K + m_e c^2, \quad E_{\text{bunch,kin}} = NK. \quad (1.2)$$

1.1 First simulation case

A straight 60 MeV, 5 nC bunch passing parallel to a flat $\epsilon_r = 2.13$ radiator is a useful first proof of concept. Varying the gap a tests coupling, spectrum, polarization and the internal Cherenkov angle before adding the tilted prism and its optical extraction. The gap, transverse beam sizes and radiator dimensions remain parameters.

A longitudinal span of 6–10 σ_z , or approximately 18–30 mm, is a starting allowance for the bunch in a laboratory-frame domain. Add the radiator, upstream and downstream vacuum margins, radiation travel distance and absorbing layers. This bunch-span estimate alone does not determine the full simulation volume or duration.

1.2 Simplified experimental considerations

The 3 mm bunch length gives a longitudinal squared-form-factor roll-off near 16 GHz, corresponding to centimetre wavelengths. This motivates examining a microwave band, but the detected maximum also depends on the radiator response and collection geometry. A flat slab validates generation inside the dielectric; the prism and optical system are needed to predict the collected signal. Section 2 separates the single-electron response, bunch coherence and finite-body effects.

The prism aperture models discussed below are most useful when face dimensions greatly exceed wavelength. A centimetre-scale radiator is comparable to the approximately 19 mm vacuum wavelength at this bunch form-factor scale, so that assumption must be checked. The appropriate material input is $\epsilon_r(\omega)$ over the chosen band; an optical dielectric constant cannot be presumed valid at microwave frequencies.

2 Analytical and reduced single-electron models

2.1 Literature closest to the stated geometry

Tyukhtin, Galyamin and Vorobev [1] give the exact dielectric-half-space field as a single transverse-wave-number integral in Section III, Eqs. (5)–(8). This is the most direct reference for the planar benchmark. The later finite-prism construction in that paper uses an aperture approximation, and should not be treated as an exact solution for any finite body.

Shevelev and Konkov [2] calculate polarization radiation from a charge outside a prismatic target, including target orientation and extraction. Their target is infinite in one transverse direction. The Fresnel extraction assumes an exit face much larger than the wavelength and omits multiple internal rereflections; see p. 510. Their gap is denoted by b , whereas their a is a target dimension.

Curcio et al. [3] connect ChDR theory, simulations and bunch-length measurements. Section III.A compares MAGIC simulations of an axisymmetric hollow cone with an analytical radiator transfer function. The modulated beam and conical geometry support the approach, but are different from a planar single-electron benchmark.

2.2 Meaning of a one-dimensional computation

For the analytical reference, replace the finite radiator in Fig. 1 by a dielectric half-space with an infinitely extended beam-facing plane at $x = 0$. Choose vacuum at $x > 0$ and dielectric at $x < 0$. Cartesian vectors below are ordered as (x, y, z) . For a prescribed electron trajectory at gap $a > 0$, moving along $+z$,

$$\mathbf{r}_e(t) = (a, 0, vt), \quad (2.1)$$

$$\rho(\mathbf{r}, t) = -e \delta(x - a) \delta(y) \delta(z - vt), \quad \mathbf{J} = v\rho \hat{\mathbf{z}}. \quad (2.2)$$

Here $v = \beta c > 0$, so $J_x = J_y = 0$ and $J_z = v\rho$. With time convention $e^{-i\omega t}$, the source fixes

$$k_z = \frac{\omega}{v}. \quad (2.3)$$

Fourier transformation in y is useful because the ideal half-space is uniform along y : spatial modes $e^{ik_y y}$ can be solved independently. Motion along z already fixes $k_z = \omega/v$. Thus, for each pair (ω, k_y) , only the dependence on x remains to be solved as a one-dimensional boundary-value problem. For $\mu_r = 1$, the vacuum decay constant and dielectric normal wave number obey

$$\kappa_0 = \sqrt{k_y^2 + \frac{\omega^2}{\gamma^2 v^2}}, \quad (2.4)$$

$$k_{x,d}^2 = \epsilon_r(\omega) \frac{\omega^2}{c^2} - \frac{\omega^2}{v^2} - k_y^2. \quad (2.5)$$

The subscript d in $k_{x,d}$ means *dielectric*: it labels the wave-number component normal to the interface inside the dielectric, not a distance or derivative. Equation (2.5) follows from the dielectric dispersion relation

$$k_{x,d}^2 + k_y^2 + k_z^2 = \epsilon_r(\omega) \frac{\omega^2}{c^2}$$

after substituting $k_z = \omega/v$. The vacuum field is evanescent normal to the interface. Choose the dielectric branch to carry energy away from the interface, or to decay into passive lossy material.

Each spectral component is obtained by matching the two polarizations at the interface. The electron is localized at $y = 0$, so its field contains all k_y components even though the material is uniform in y . Integrating these components over k_y reconstructs the three-dimensional point-electron field at a fixed frequency. Reconstructing a time waveform also requires an inverse frequency transform. Both propagating and evanescent dielectric components belong in the full integral.

This is a *spectral reduction of a three-dimensional point-source problem*. Ordinary 1D FDTD cannot resolve both the gap and the propagation direction. Ordinary Cartesian 2D FDTD describes a source uniform in the omitted coordinate, hence a line charge with per-unit-length normalization. The explicit amplitudes and Fourier normalization in Ref. [1] use Gaussian electromagnetic units; convert source and field normalization consistently for an absolute comparison in SI or IPPL units.

2.3 Threshold, internal angle and gap dependence

For a real, positive, nondispersive permittivity, a propagating dielectric component requires

$$k_y^2 < \frac{\omega^2}{c^2} (\epsilon_r - \beta^{-2}). \quad (2.6)$$

Consequently the threshold and internal phase-matching angle are

$$n\beta > 1, \quad \cos \theta_{\text{Ch}} = \frac{1}{n\beta}, \quad \theta_{\text{Ch}} = 46.7476^\circ \quad (\epsilon_r = 2.13). \quad (2.7)$$

This angle is measured inside the homogeneous dielectric relative to the beam direction $+z$. It does not specify the camera angle after extraction.

At a fixed k_y , moving the trajectory away from the interface supplies the amplitude factor $e^{-\kappa_0 a}$, and the corresponding modal energy factor $e^{-2\kappa_0 a}$. An integrated observable combines many k_y values and need not follow a single exponential in a . At $k_y = 0$,

$$\kappa_0^{-1} = \frac{\gamma\beta\lambda_0}{2\pi}. \quad (2.8)$$

For example, this field coupling length is approximately $9.4 \mu\text{m}$ at a vacuum wavelength of 500 nm . It is neither a hard cutoff nor a required box dimension.

2.4 Extension to the specified bunch

For prescribed trajectories in a linear material, electron responses can be superposed. If they differ only by independent arrival times and have the same transverse coupling, expanding the squared sum of the amplitudes gives

$$\left\langle \frac{dW_N}{d\omega} \right\rangle = [N + N(N-1)|F(\omega)|^2] \frac{dW_1}{d\omega}, \quad (2.9)$$

$$|F(\omega)|^2 = \exp[-(\omega\sigma_t)^2] = \exp[-(\omega\sigma_z/v)^2], \quad N = \frac{Q}{e}. \quad (2.10)$$

A transverse distribution, including a distribution of gaps, requires an amplitude average over those trajectories.

The Gaussian squared form factor reaches e^{-1} at

$$f_* = \frac{1}{2\pi\sigma_t} = \frac{v}{2\pi\sigma_z} = 15.904 \text{ GHz}, \quad \lambda_{0,*} \simeq 18.85 \text{ mm}. \quad (2.11)$$

This is a form-factor roll-off, *not automatically a detected spectral maximum*. The single-electron spectrum, finite-radiator response and collection optics also contribute. A direct representation of individual electrons does not remove the grid's spatial and temporal bandwidth limits. For a fixed bunch shape and geometry, the coherent contribution scales approximately as Q^2 . At visible wavelengths the Gaussian longitudinal coherent term is strongly suppressed for a 3 mm bunch. A visible-camera prediction therefore needs the incoherent single-electron contribution, or optical-scale bunching if coherent emission is intended.

2.5 Finite radiator and the approximately 47-degree geometry

Equation (2.3) implies $k_z > \omega/c$: the tangential wave number exceeds the vacuum propagating-wave limit. Therefore a mode coupled through the infinite planar interface cannot escape as propagating radiation through that same interface. An inclined extraction face or a finite edge changes this constraint. This is a phase-matching consequence of the idealized model.

A rectangular radiator can demonstrate excitation inside the dielectric, while producing a different external angular distribution from a prism. Distinguish the orientation of the beam-facing surface, the extraction-face normal and the internal angle in Eq. (2.7).

Entering and leaving the region beside a finite body breaks translation symmetry along z . Edge transients, interference and internal reflections then require a finite-geometry treatment. If the material is still invariant in y , a Fourier transform in that coordinate gives a family of two-dimensional problems, often called 2.5D. Finite width generally requires a three-dimensional calculation.

3 FDTD and the vacuum–dielectric boundary

3.1 Field equations and the material interface

The Meep paper by Oskooi et al. [4] describes a practical FDTD implementation and includes moving-source Cherenkov radiation in Fig. 5. Its homogeneous-medium, two-dimensional demonstration is a useful starting point. The maintained source example is also available [5].

A suitable material formulation, written here in SI units, is

$$\partial_t \mathbf{D} = \nabla \times \mathbf{H} - \mathbf{J}_{\text{free}}, \quad \partial_t \mathbf{B} = -\nabla \times \mathbf{E}, \quad (3.1)$$

$$\mathbf{B} = \mu_0 \mathbf{H}, \quad \mathbf{D} = \epsilon_0 \epsilon_r(\mathbf{r}) \mathbf{E}. \quad (3.2)$$

For the initial lossless, nondispersive proof of concept, $\epsilon_r = 1$ in vacuum and 2.13 in the radiator. The electron’s field polarizes the material; a separate prescribed Cherenkov-radiation source is unnecessary.

With no free surface-charge sheet or free surface current, the physical interface conditions are

$$[\mathbf{E}_t] = [\mathbf{H}_t] = 0, \quad [D_n] = [B_n] = 0. \quad (3.3)$$

Here square brackets denote the jump across the interface. The normal components at the beam-facing plane $x = 0$ are D_x, B_x , and the tangential components are E_y, E_z, H_y, H_z . The normal electric field generally jumps; forcing every component of $\mathbf{E}$ to be continuous gives an incorrect boundary. Bound polarization charge is represented by the constitutive relation.

In a Yee discretization, material coefficients must correspond to the electric-component locations and their averaging volumes. Beginning with a grid-aligned plane isolates constitutive and source errors from errors in representing a sloping face.

3.2 Cut cells and a tilted extraction face

Farjadpour et al. [6] derive subpixel averaging that removes the leading interface error for suitable geometries. Kottke et al. [7], Section VI, explain the discontinuity-aware treatment and its anisotropic extension.

For a cell cut by a locally planar interface between two isotropic, nondispersive materials, let f be the dielectric fraction. Using relative permittivities, the tangential and normal averages are

$$\epsilon_{\parallel} = f\epsilon_d + (1 - f)\epsilon_v, \quad (3.4)$$

$$\epsilon_{\perp} = \left(\frac{f}{\epsilon_d} + \frac{1 - f}{\epsilon_v} \right)^{-1}. \quad (3.5)$$

For unit interface normal $\hat{\mathbf{n}}$, these form the tensor

$$\epsilon_{\text{eff}} = \epsilon_{\parallel}(\mathbf{I} - \hat{\mathbf{n}}\hat{\mathbf{n}}^T) + \epsilon_{\perp}\hat{\mathbf{n}}\hat{\mathbf{n}}^T. \quad (3.6)$$

At the beam-facing plane, $\hat{\mathbf{n}} = \hat{\mathbf{x}}$, so in (x, y, z) order this is $\text{diag}(\epsilon_{\perp}, \epsilon_{\parallel}, \epsilon_{\parallel})$. Off-diagonal entries require consistent interpolation on a staggered grid. This describes a sharp interface within a numerical cell, rather than a deliberately thick physical transition layer. Corners and edges can limit convergence below second order [7].

A 47° face is therefore possible on a Cartesian grid. The issue is accurate representation and convergence of the interface. Arbitrary scalar averaging of the permittivity does not implement Eq. (3.6).

3.3 Frequency-dependent response

For dispersive material, introduce a causal polarization model, for example auxiliary equations for Lorentz or Debye responses, with

$$\mathbf{D} = \epsilon_0 \epsilon_\infty \mathbf{E} + \mathbf{P}. \quad (3.7)$$

The nondispersive averaging above is not a complete dispersive algorithm. Meep smooths the instantaneous permittivity and documents limitations for dispersive contributions [8]. Material loss, bandwidth and any auxiliary-equation stability conditions need explicit choices.

3.4 Moving charge, initialization and conservation

Esirkepov's deposition construction [9] provides a reference for satisfying discrete continuity. Its discrete operators must match those of the selected Maxwell solver:

$$\frac{\rho^{n+1} - \rho^n}{\Delta t} + \nabla_h \cdot \mathbf{J}^{n+1/2} = 0. \quad (3.8)$$

Combined with consistent initial fields, this preserves the free-charge Gauss constraint $\nabla_h \cdot \mathbf{D} = \rho$. Starting an already moving electron with zero fields, or abruptly switching its current on and off, creates transients. Initialize consistently or show that those transients do not enter the benchmark measurement.

The numerical particle shape must fit within the vacuum gap and resolve the measured frequencies. Otherwise deposition overlaps the material or filters the spectrum. These requirements still apply when every simulated particle represents one electron.

3.5 Outer absorption and numerical radiation

A perfectly matched layer (PML) truncates the computational domain; it is separate from the physical vacuum–dielectric interface. Johnson's notes [10] explain the construction and its limitations. Place it outside the observation region, and verify convergence with its distance and thickness. If dielectric reaches the PML, extend the material consistently. Inhomogeneity along the stretching direction, including some oblique structures, can invalidate ordinary matching [11].

The setup draft additionally cites Kraus, Adelmann and Arbenz [12], who formulate PML for a divergence-preserving alternating-direction implicit (ADI) Maxwell scheme. This is relevant when evaluating an ADI route; its update equations must not be assumed to apply unchanged to the present explicit potential solver.

Xu et al. [13] analyse spurious fields and Cherenkov-like wakes around a relativistic particle, including in vacuum. At the present $\beta = 0.99996434$, a vacuum control with the same source, mesh and time step is essential. A visible cone alone is not evidence of physical ChDR. Numerical radiation from a prescribed charge should also be distinguished from collective numerical Cherenkov instability.

Measure radiation separately from the electron's bound near field: compare complex field spectra at specified locations, or outgoing flux away from the trajectory. Align $\mathbf{E}$ and $\mathbf{H}$ in space and time before computing Poynting flux [4]. Subtracting two near-field powers is not generally equivalent to computing the power of a difference field.

4 Implications for the new IPPL ChDR mini-app

4.1 Development scope for the MSc thesis of D. Rosser

The setup draft identifies dielectric interfaces, the reference frame and absorbing boundaries as development topics for this thesis. The following items connect that scope to the current implementation.

Field formulation. `demos/fel/datatypes.h`, around line 50, selects `NonStandardFDTD Solver` with absorbing Mur boundaries. `src/MaxwellSolvers/NonStandardFDTD Solver.hpp`, around lines 28

and 114, advances four-potential fields using a vacuum stencil and sets $\Delta t = h_z$ in normalized units. A spatial dielectric is more than a geometry mask or a replacement of c by c/n : constitutive response, interface coupling and stability must be derived consistently.

Existing particle and source machinery. The FEL mini-app already provides relativistic Boris particle motion, field interpolation, particle initialization and current deposition. In `demos/fel/FreeElectronLaserManager.h`, `depositCurrent` calls `assemble_current_collocated`; CIC charge-density deposition is enabled by `space_charge`. These are existing capabilities to reuse for ChDR. The material extension does not by itself require a new particle pusher or deposition scheme.

Initialization and conservation checks. Section 3.4 is therefore supported by existing particle/source machinery, but its full consistency requirements have not yet been established for the ChDR configuration. In `src/MaxwellSolvers/NonStandardFDTDSolver.hpp`, `initialize` sets all three potential time levels to zero. The manager's initialization sequence does not supply the pre-existing self-field of an already moving electron. Consistent field initialization, or exclusion of startup transients from the measurement, remains necessary.

The current implementation uses trajectory-segmented CIC deposition on a collocated grid. The tests in `unit_tests/Interpolation/TestCurrentDeposition.cpp` check deposited values, but do not establish the discrete continuity relation in Eq. (3.8) or preservation of the Gauss constraint. Exact discrete charge conservation should therefore be verified using the selected field operators; it is not established by the presence of the deposition routine alone. Repeat that compatibility check if the material solver or field staggering changes.

Moving electron source: Oskooi and Johnson's chapter in the volume edited by Taflové, Oskooi and Johnson [14, 15], Section 4.7, pp. 89–91, explicitly demonstrates moving-source Cherenkov radiation. Figure 4.10 compares continuous interpolation of the source with rounding its position to grid points; the latter produces artificial high-frequency radiation. Interpolation reduces these artifacts, while numerical dispersion remains. This homogeneous-medium, two-dimensional example supports source validation but does not replace the vacuum-gap benchmark for our geometry. The chapter is authored by Oskooi and Johnson; Taflové is a co-editor of the book. Taflové and Hagness [16] provide the broader FDTD background on material updates, numerical dispersion, stability and absorbing boundaries.

Reference frame. The same manager, around line 47, ties a Lorentz boost to bunch and undulator parameters. A laboratory-frame ChDR mode keeps the radiator stationary. A dielectric that is stationary and isotropic in the laboratory cannot be represented as the same stationary scalar material after a nonzero boost. Removing the undulator field alone does not establish the correct material-frame model.

Outer boundaries. The second-order Mur treatment can be used for preliminary checks if reflections are shown to be below the measurement tolerance. Quantitative power and oblique radiation motivate a validated PML implementation. The absorber must remain separate from the physical dielectric interface.

Two possible formulations merit evaluation:

1. A direct $\mathbf{D}, \mathbf{B}$ Maxwell solver with a material constitutive response and consistent particle coupling.
2. A potential solver coupled self-consistently to polarization charge and current,

$$\rho_{\text{bound}} = -\nabla \cdot \mathbf{P}, \quad \mathbf{J}_{\text{bound}} = \partial_t \mathbf{P}. \quad (4.1)$$

This route needs a stable, gauge-consistent coupling.

The literature review does not establish which requires the smaller implementation change.

4.2 Job-file units and coordinate mapping

The parser in `demos/fel/Config.h` converts the numerical charge and mass entries from elementary-charge and electron-mass units into solver units. They are not direct SI coulombs and kilograms. The manager subsequently divides the configured totals by the actual particle count.

The charge conversion factor in `demos/fel/units.h` is positive and the input sign is passed through. Thus, for an electron bunch of charge -5 nC, the setup draft's charge magnitude becomes a negative input. A sketch of the relevant entries is

```
"charge": -3.1207545e10,
"mass":    3.1207545e10,
"gamma":   118.417071,
"sigma-position": [sigma_x_code_um, sigma_y_code_um, 3000.0]
```

The two transverse sizes are placeholders. The value 3000.0 assumes the job file selects micrometres as its length scale. In the current code the third coordinate, z_{code} , is longitudinal and corresponds to z in Fig. 1; x_{code} and y_{code} are transverse. Thus the third entry is $\sigma_z = 3000 \mu\text{m}$, while the first two are σ_x, σ_y . This coordinate convention does not remove the code's boost. These entries therefore specify units and beam scales, not a complete runnable laboratory-frame ChDR job.

4.3 Validation sequence

1. **Passive interface.** Launch plane waves at a flat boundary. Compare Fresnel reflection and transmission for both polarizations and check lossless energy balance.
2. **Vacuum electron.** Use a prescribed 60 MeV trajectory and consistent initial fields. Measure unwanted radiation, continuity residuals and the Gauss residual.
3. **Planar ChDR.** Compare complex spectra, polarization, gap dependence and propagation angle with the independent half-space calculation. Check convergence in mesh spacing, particle shape, box dimensions and outer absorption.
4. **Finite radiator.** Add leading and trailing edges, followed by the tilted extraction face. Compare angular spectra and refine the interface representation. Apply finite-target analytical approximations only within their stated regime.
5. **Bunch response.** Introduce the longitudinal and transverse distributions in the selected band after the single-electron response is established.

For an infinite track, compare a spectral energy quantity per unit path length rather than a finite total radiated energy. For a finite radiator, define whether the observable is a local field, electron energy loss, outgoing spectral energy or energy collected by specified optics.

For a conventional explicit Yee solver, the vacuum region imposes

$$c\Delta t \sqrt{h_x^{-2} + h_y^{-2} + h_z^{-2}} \leq 1. \quad (4.2)$$

Auxiliary material equations and other stencils can impose additional conditions. This is not the time-step rule of the existing nonstandard potential solver. Resolve the shortest material wavelength and relevant gap/source scales; a prescribed cell count alone does not certify accuracy.

4.4 Inputs still needed and status

The remaining physical choices are gap a , radiator length, thickness and width, face orientations, material data $\epsilon_r(\omega)$, the frequency band, transverse beam distribution and the detector observable. The approximately 16 GHz bunch form-factor scale is useful for planning, but does not determine the detected peak or eliminate the need for the radiator transfer function.

This is a focused literature synthesis and implementation assessment. The beam quantities are direct calculations from the stated assumptions. The listed simulations and convergence checks are recommendations; no field simulation or solver modification was performed for this report.

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